
A NOVEL MEDICAL IMAGE SEGMENTATION METHOD VIA MASKED PRETRAINING

A PREPRINT


Department of Molecular Medicine
Aarhus University


Department of Computer Science
Aarhus University

December 5, 2025

ABSTRACT

In the field of medical imaging, collecting large-scale pixel-wise annotations for panoramic X-rays is costly and time-consuming, while a large portion of the available data remain unlabeled. We propose and evaluate a novel semi-supervised medical image segmentation framework that leverages masked self-supervised pretraining to enhance the performance of UNet models when only limited labeled images but abundant unlabeled data are available. During pretraining, random subsets of input pixels are masked, and the UNet model is trained to reconstruct the missing pixels and then is fine-tuned for the supervised segmentation task. In addition, we test NAC-UNet, a lightweight variant of UNet tailored for the ultrasound image segmentation task.

Experiments are conducted on the TG3K ultrasound dataset of thyroid nodules, and STS-Tooth 2D panoramic X-ray image dataset, which consists of 4,000 unlabeled and 900 labeled samples. Experimental results show that the proposed masked pretraining improves segmentation accuracy, yielding a 2.95% improvement in Dice score on TG3K dataset over purely supervised training. Moreover, scAG-UNet achieves competitive performance compared to the standard U-Net.

Keywords Deep Learning · Self-Supervised Learning · Masked Image Modeling · Medical Image Segmentation · UNet

1 Introduction

Panoramic X-ray and ultrasound images are widely used in medical practice for diagnosis and treatment planning. Precise segmentation at the pixel-level in these images can serve as a foundation for downstream detection or classification systems. [Rubiu et al. \[2023\]](#) However, annotating such images is costly and time-consuming. [Wang et al. \[2025\]](#)

To reduce reliance on fully labeled data, masked image modeling (MIM) has become an important method in computer vision. Masked image modeling can learn the latent representations of large-scale unlabeled images and later transfer these representations to downstream tasks with limited labels. [VanBerlo et al. \[2024\]](#) In particular, masked autoencoder-style approaches that reconstruct missing regions of an image from the visible context have recently shown strong performance for vision transformers and other architectures, for it encourages the model to learn holistic, high-level features rather than simple low-level extrapolation. [He et al. \[2022\]](#)

Furthermore, unlike token-based Transformer models, UNet operates in the pixel domain and is more computationally efficient. However, the working principle behind MIM is not well explained, and previous studies insist that MIM primarily works for the Transformer family but is incompatible with CNN-based models.

Therefore, we aim to investigate the extent to which UNet can benefit from masked image pretraining and identify the key factors that influence its downstream segmentation performance.

2 Related Work

2.1 Precise image segmentation with limited labels

Image segmentation requires the highest level of supervision, making it highly sensitive to annotation quality and quantity. Obtaining a large amount of annotated data is time-consuming and expensive, especially in medical imaging where expert annotation is needed and datasets are limited. Researchers have explored various strategies to mitigate the constraints by limited data from simple data augmentation and regularization to more advanced deep learning methods.

Transfer learning allows models to leverage knowledge learned from large pretrained datasets and apply them to downstream tasks, thereby enabling faster and more efficient training on limited data. However, this approach comes with challenges. Main limitations include domain mismatch and overfitting on small datasets [Alzubaidi et al., 2021]. Domain shifts are particularly obvious in medical imaging, where feature distributions vary across different clinical modalities. Addressing these gaps is essential for effective adaptation of pre-trained models in medical imaging.

Semi-supervised learning leverages both labeled and unlabeled data using methods like pseudo-labeling and consistency regularization. Its performance, however, is highly dependent on pseudo-label quality and dataset balance. To enhance model performance, several studies have introduced strategies for improving pseudo-label reliability, including the incorporation of similarity matrices [Liu et al., 2024], the investigation of pseudo-label thresholding schemes [Zeng et al., 2023], and the assessment of model-derived confidence measures. Despite these advancements, the additional computational overhead remains a substantial limitation.

Self-supervised learning (SSL) utilizes large-scale unlabeled datasets via a two-stage process: pretraining on a certain task to learn latent features, followed by fine-tuning on limited labeled data. Recent studies have demonstrated the efficacy of SSL in medical imaging segmentation [Felfeliyan et al., 2023], where labeled datasets are often scarce. However, the performance of SSL heavily depends on the careful selection of an appropriate pretraining task. Furthermore, the two-stage training process poses challenges, as it requires significant computational resources and extended training time.

2.2 Masked image pretraining

Masked image pretraining is a self-supervised paradigm that learns visual representations by reconstructing missing content from partially observed images. Given an input image $x \in \mathbb{R}^{C \times H \times W}$ and a binary mask $m \in \{0, 1\}^{H \times W}$ (where $m_{ij} = 1$ denotes a masked pixel), the encoder f_θ only sees the corrupted input $(1 - m) \odot x$, and a decoder g_ϕ is trained to reconstruct the full image. We optimise a reconstruction loss restricted to the masked regions,

$$\mathcal{L}_{MIM} = \|(g_\phi(f_\theta((1 - m) \odot x)) - x) \odot m\|_1, \quad (1)$$

so that the encoder must infer occluded structures from surrounding context instead of relying on trivial low-level cues. Masked Autoencoders (MAE) show that a simple pixel-regression objective with very high random masking ratios can scale efficiently and yield strong representations for Vision Transformers [He et al., 2022], while SimMIM demonstrates that even a minimal design with a lightweight prediction head and uniform random masks performs competitively on ImageNet and downstream tasks [Xie et al., 2022].

Beyond classification, masked image pretraining has been shown to transfer well to dense prediction. BEiT formulates masked image modeling as a discrete token prediction problem and reports competitive semantic segmentation results on ADE20K when fine-tuning masked-pretrained Transformers, compared with supervised ImageNet pretraining [Bao et al., 2022]. iBOT further combines masked prediction with self-distillation to emphasise local semantics and achieves state-of-the-art performance on object detection, instance segmentation and semantic segmentation using plain ViT backbones [Zhou et al., 2022].

These results suggest that forcing the encoder to reconstruct hidden regions encourages it to capture boundaries, spatial layout and long-range context, which are precisely the properties needed for accurate region-level delineation in segmentation. However, existing MIM methods are largely designed for patch-based Transformer architectures trained on large-scale natural image corpora, whereas clinical segmentation pipelines still predominantly rely on convolutional encoder-decoder networks such as UNet, especially for grayscale modalities like panoramic dental X-rays and ultrasound. It is therefore not obvious that the gains observed for Transformer-based MIM will directly transfer to UNet-style models under severe label constraints. In this work, we adopt the simple pixel-regression objective in Eq. 1 with random spatial masks and investigate how masked image pretraining affects UNet-like architectures for dense prediction, by pretraining on large collections of unlabeled dental and thyroid ultrasound images and subsequently fine-tuning on limited pixel-wise annotations.

3 Methods

3.1 Dataset

Our primary experiments are carried out on the TG3K thyroid ultrasound dataset, with supplementary experiments on the STS-2D-Tooth panoramic dental X-ray dataset.

TG3K thyroid ultrasound. TG3K comprises 3,583 B-mode ultrasound frames extracted from 16 cine sequences of the thyroid gland, each accompanied by a pixel-wise binary mask of the gland region. The frames were curated such that the gland occupies at least 6% of the image area, providing reliable delineations for supervised learning. All images are single-channel and are resampled to 224×320 pixels prior to training. For the segmentation task, we partition the dataset into training, validation and test splits, following the official ID lists. To mimic a limited-annotation scenario, we restrict the number of labeled training images to 500 and treat the remaining training frames as unlabeled during pretraining. In the masked image modeling stage, only the raw ultrasound images are used and all segmentation masks are ignored; the held-out labels are used exclusively for downstream fine-tuning and evaluation.

STS-2D-Tooth panoramic X-rays. STS-2D-Tooth contains 4,000 panoramic dental X-ray images (PXI), including 3,500 adult and 500 child scans, of which 900 images are provided with pixel-level tooth-region masks [Wang et al., 2025]. The remaining 3,100 images are unlabeled and are intended for semi-supervised or self-supervised learning. In this work we only use the 2D PXI subset. All images are converted to grayscale, resized to 224×320 pixels, and normalised using the same procedure as for TG3K to enable a shared model architecture. For supervised segmentation on STS-2D-Tooth, we operate on the 900 labeled images and divide them into training, validation, and test sets with an 80/10/10 split. For masked pretraining, we combine all 4,000 PXI images and ignore the segmentation masks, so that the self-supervised objective is defined solely in terms of reconstructing masked pixels.

Both datasets provide pre-defined patient-level annotations and train/validation/test splits, ensuring that samples from the same patient do not appear across different subsets during training.

3.2 Data preprocessing and data enhancement

For both datasets we standardize the input so that it matches the assumptions of our UNet-based models and reduces variability across acquisitions. All images are converted to single-channel grayscale and resampled to a fixed resolution of 224×320 pixels, after which intensities are linearly scaled to $[0, 1]$ and normalized using a global mean of 0.5 and standard deviation of 0.25 estimated from the TG3K training set. During supervised training on TG3K we apply light geometric augmentation jointly to image and mask, consisting of random horizontal flips and small in-plane rotations and shifts, in order to increase the effective dataset size while preserving the anatomical plausibility of the thyroid gland. For masked image pretraining we use the same normalisation and geometric transforms and additionally generate random block-wise masks that occlude a fixed fraction of pixels; the masked image is used as the network input, and the corresponding uncorrupted image serves as the reconstruction target. No random augmentation is applied to validation data, where we only perform resizing and normalisation.

3.3 Model architecture

All models follow a 2D encoder-decoder design with skip connections. The input is a single-channel image of size 224×320 , and the output is a single-channel logits map for foreground-background segmentation. The encoder consists of a sequence of downsampling blocks, each formed by a DoubleConv module (two 3×3 convolutions with batch normalisation and ReLU) followed by 2×2 max pooling. The decoder mirrors this structure with upsampling blocks that use bilinear upsampling and convolution, concatenate the corresponding encoder feature map, and apply another DoubleConv. A final 1×1 convolution produces the segmentation logits. Channel widths increase from 32 up to 512 in the encoder and decrease symmetrically in the decoder.

On top of this baseline UNet, we consider two variants. The first, denoted UNet2D-scAG [Khanh et al., 2020], augments each skip connection with a spatial-channel attention gate (scAG). For a given encoder feature map F_e and decoder gating signal F_g , scAG computes channel-wise and spatial attention maps and reweights the encoder features before concatenation. This allows the decoder to suppress irrelevant background responses and emphasise structures that are consistent with the current decoding stage, which is particularly useful in noisy ultrasound and X-ray images.

The second variant, UNet2D-NAC [Tao et al., 2022], inserts a neighbour-aware context (NAC) block at the bottleneck. NAC consists of four parallel 3×3 dilated convolutions, flexibly adjusting the appropriate convolution according to the size of the nodule. The fused feature map provides multi-scale context without adding many parameters. UNet2D-NAC also retains the scAG modules in the decoder, and is used as our main “enhanced” architecture in the experiments.

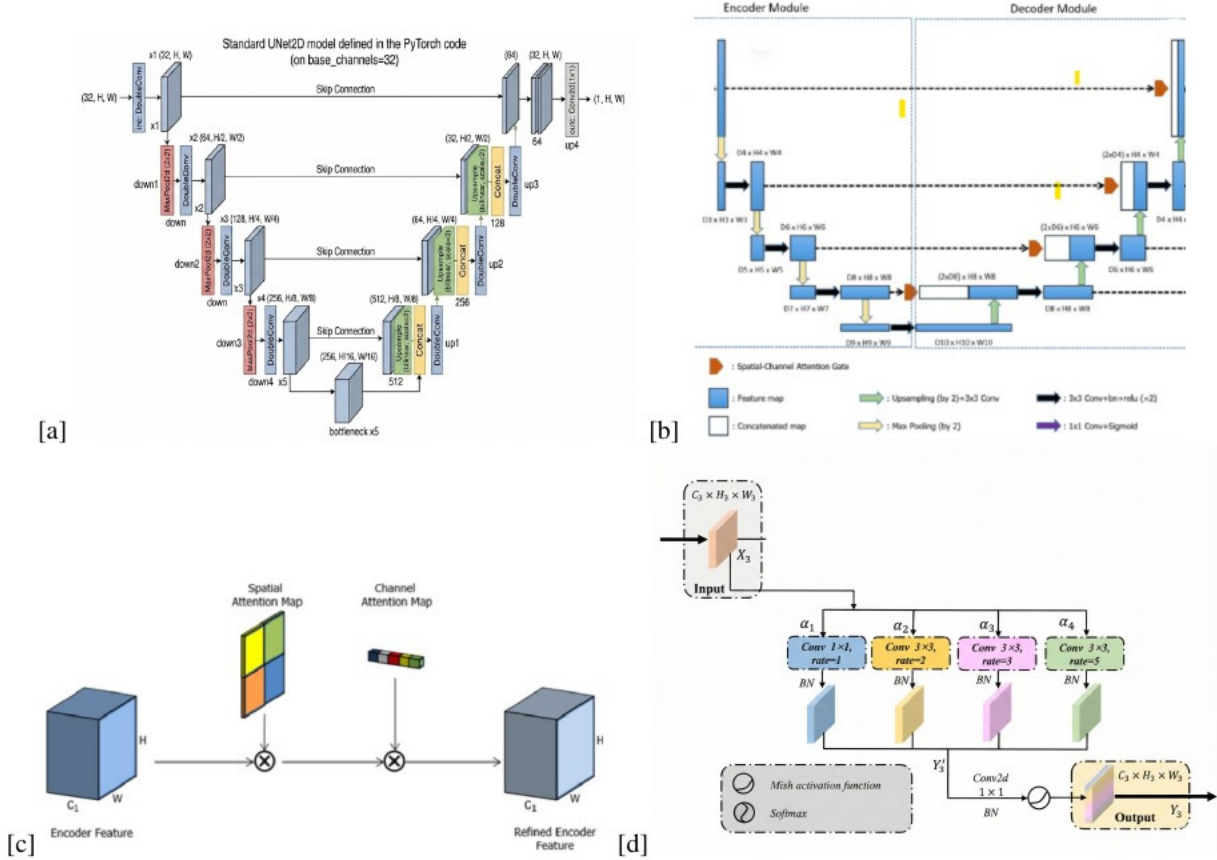


Figure 1: UNet baseline and its variants

Figure 2: Overview of the model architectures used in this work. (a) Baseline 2D UNet with encoder–decoder and skip connections, (b) Overall structure of scAG-UNet, (c) Spatial–channel attention gate (scAG) module that refines encoder features before fusion and (d) Neighbour-aware context (NAC) module that aggregates multi-scale contextual information at the bottleneck.

3.4 Masked pretraining approach

The masked pretraining approach is a simple autoencoder-like approach that tries to reconstruct the pixels of the original image given the partial observation (masked image), as shown in Figure 3. Unlike the current pretraining methods, the

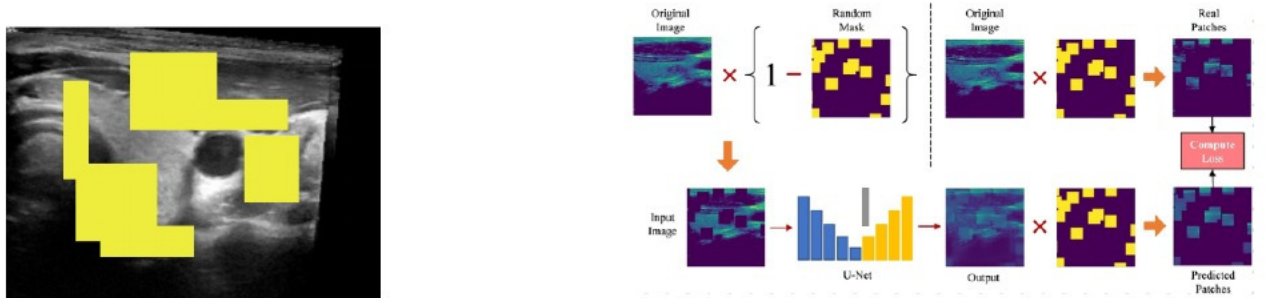


Figure 3: The masked pretraining architecture. A small proportion of pixels in the original image are selected by random patches and masked out, and the UNet model is applied to the remaining pixels to reconstruct the original image in pixels. The training loss is computed by comparing the real and predicted pixels in the masked area.

UNet and its variants are used as backbone during both pretraining and supervised segmentation. The model doesn't explicitly learn a latent representation of the large-scale images. Some existing pretraining methods reuse only part

of the model such as the encoder in the fine-tuning stage. We believe that the decoder and skip connection are useful contain high-level semantic information, so for the segmentation, the parameters in both the encoder and decoder obtained by pretraining are used as the initial state. We apply a patch-based random masking strategy, which uses rectangular regions of varying sizes and locations, at a fixed ratio of the masked area to the entire image. Specifically, the mask is generated by repeatedly placing random square or rectangular blocks—each sized between *max_block_size* and *min_block_size* onto the image until the desired masked area.

3.5 Supervised segmentation

In supervised segmentation a model learns to label each pixel of an image from known image–mask pairs. Because we already have initialized weights from the mask pretraining, and those weights are learned using inputs scaled to the 0–1 range (compatible with the binary mask), we can reuse the same model to conduct supervised training.

4 Experiments

4.1 Metrics

We use the Dice similarity coefficient as the primary evaluation metric. Given the ground-truth mask G and the predicted binary mask P , Dice is defined as

$$\text{Dice}(G, P) = \frac{2|G \cap P|}{|G| + |P|},$$

where G and P denote the sets of foreground pixels in the ground truth and prediction, respectively. In practice, we obtain P by applying a sigmoid to the network logits and thresholding at 0.5.

4.2 Loss function

For the reconstruction task for pretraining, we use L_1 reconstruction loss computed only over masked pixels.

For the supervised segmentation stage, we define a combined segmentation loss that balances pixel-wise accuracy and region-level overlap. The loss is defined as:

$$\mathcal{L} = \text{BCE}(y, \hat{y}) + \lambda \mathcal{L}_{\text{Dice}}(y, \hat{y})$$

The symbol λ is a hyperparameter controlling the relative weight of the Dice loss in the combined segmentation loss.

4.3 Hyperparameters

We control several hyperparameters in our experiments. The base number of feature maps is set via *base_channels*. Input images were resized to a uniform height and width (*img_height* and *img_width*) to handle heterogeneous image dimensions. Training is conducted with a batch size of 4 for 50 epochs, using *learning_rate* and *weight_decay*; data loading parallelism is controlled by *num_workers*. For U-Net upsampling, bilinear interpolation is used unless *no_bilinear* is specified. In the masked image modeling pretraining, the fraction of masked pixels is controlled by *mask_ratio*, and the size of each masked block is constrained between *min_block_size* and *max_block_size*.

4.4 Optimization

For our optimization algorithm we use Adaptive Momentum Estimation (Adam). Weight decay is also applied as a regularization to prevent overfitting. Grid search over the Learning rate and weight decay space is performed to find the optimal parameters.

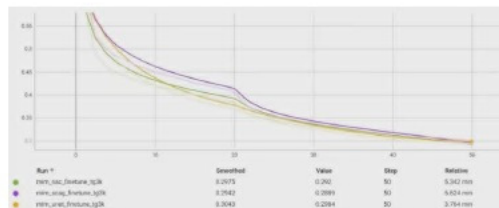


Figure 4: Loss curve

5 Results

5.1 Masking experiments

This subsection shows the impact of different masking strategies on the masked pretraining. In our proposed masking method, the *max_block_size* and *min_block_size* are set to 32 and 96, imposing constraints on the mask shape. UNet is chosen as the base model and TG3K as the benchmark. The visual comparison of reconstruction by MS-UNet pretrained with different mask ratios are shown in Figure 5

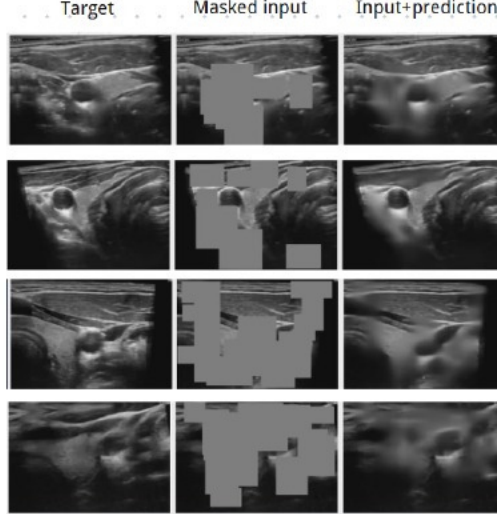


Figure 5: The reconstruction effect of pretrained models with different masking ratios (0.3, 0.3, 0.5, 0.5 from top to bottom)

We investigate the effect of different masking ratios in pretraining on the segmentation performance, as shown in Table 1

Besides, we also testing different losses in pretraining, such as restoring all the pixels in the masked and unmasked areas. It can be seen that the masked pretraining largely improves the segmentation performance with a limited number of labeled data, and our random masking strategy is basically compatible with applying full image reconstruction during pretraining, while avoiding extra computation cost.

Mask ratio	Dice(%) for val	Dice(%) for test
No pretraining	67.91	67.15
0.3	70.23	69.12
0.5	70.86	69.64
0.7	69.52	68.31
0.3(full img recon)	70.62	69.54
0.5(full img recon)	69.82	69.15
0.7(full img recon)	69.78	69.03

Table 1: The impact of the masking ratio and pretraining strategies in pretraining on the segmentation performance

5.2 Model comparison

After conducting the mask test, we are now interested in which variants of UNet performs the best in our task. We fix the mask ratio to 0.5 and set TG3K as the benchmark.

Since all three models share the same UNet backbone, for fair comparison, we specify the same number and latent dimensions of both encoder and decoder.

Figure 6 illustrates the examples of prediction by different UNet variants:

Model	Dice(%) for val	Dice(%) for test
UNet	70.86	69.64
scAG-UNet	74.09	73.12
NAC-UNet	69.67	68.55

Table 2: Results on TG3K with different model architecture

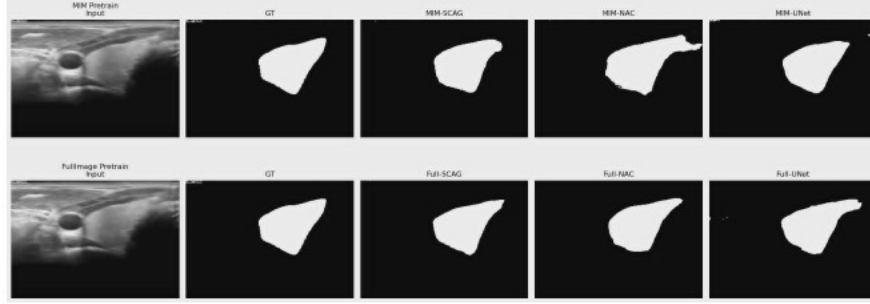


Figure 6: Segmentation examples

It can be seen from the Table 2 that the introduced attention module in scAG-UNet exhibited notably improved performance. In contrast, the NAC-UNet specifically designed for flexible nodule sizes performs even worse than the baseline UNet model. Our explanation is that the additional NAC module may increase the model complexity relative to the limited size of our dataset, which results in training instability without careful tuning of the hyperparameters.

6 Discussion

In this section we will show what strategies and tricks we tried along the way and multiple factors we discovered that impact the final segmentation performance.

6.1 Effect of dataset property

After training our segmentation model on the TG3K dataset, we also test the ability of our proposed method on STS-Tooth 2D dataset with 20% labeled images. The masked pretraining only yields only a small increase in test dice

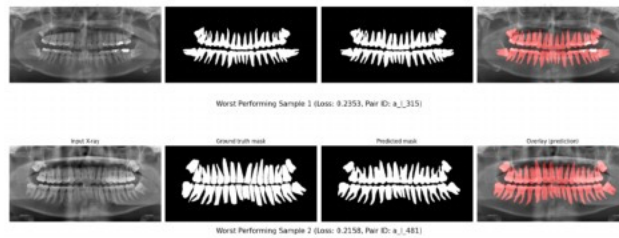


Figure 7: Top-2 worst cases of masked image pretraining on STS-Tooth-2D image segmentation

score, from 89.72% from to 90.14%, which is considered negligible and causes computational overhead. Even the worst cases for this dataset remain acceptable, as shown in Figure 7. Our explanation is that, for images with homogeneous property and smooth intensity transitions, the segmentation task is low in complexity, while the masked pretraining helps more when the dataset has complex textures and heterogeneous structures.

6.2 Effect of data preprocessing

As for the data preprocessing for the TG3K dataset, to handle the issue that images have different shape, all the images are resized to a uniform shape, which resulted in some loss of resolution and the introduction of black padding regions in certain cases. This preprocessing step may explain the performance bottleneck observed in our models.

6.3 Data augmentation issue

To address the data sparsity and increase sample diversity, we initially explored several data augmentation methods, such as Gaussian-based perturbations. However, our experiment indicates that these methods could degrade the segmentation performance. Consequently, in the final experiment setup, we restrict augmentation to horizontal flips and small rotations.

6.4 Qualitative analyses for the model learned

We would like to understand what our model actually learn and when does it succeed or fail. Below in Figure 8 are our top best and worst segmentation in terms of the validation loss: We find out that segmentation failures often occur when

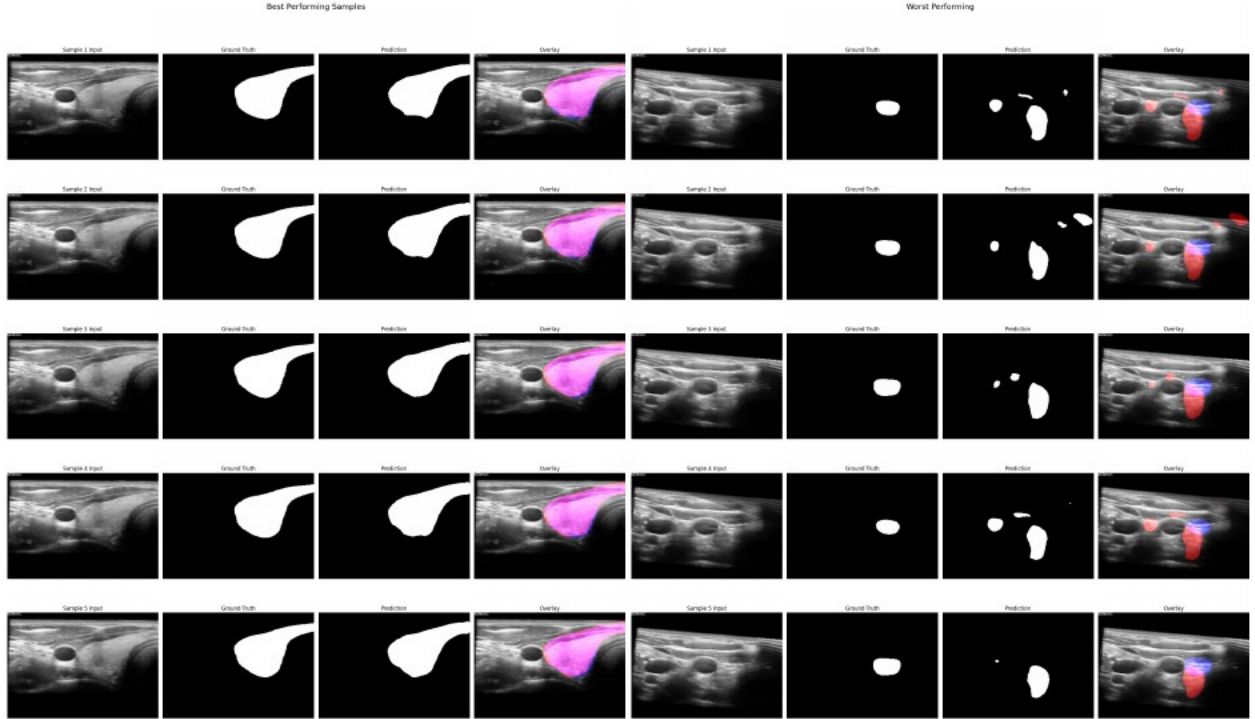


Figure 8: Top-5 best and worst predicted mask

the ground truth mask is small in size, and is surrounded by misleading or noisy pixel-value fluctuations.

Furthermore, we apply interpretable deep learning methods to investigate whether the model focuses on the features we intuitively expect. We use saliency map to visualize the pixels that to what extent they influence a model’s prediction, as shown in Figure 9. We observe that poor segmentation outcomes often coincide with the model attending to regions

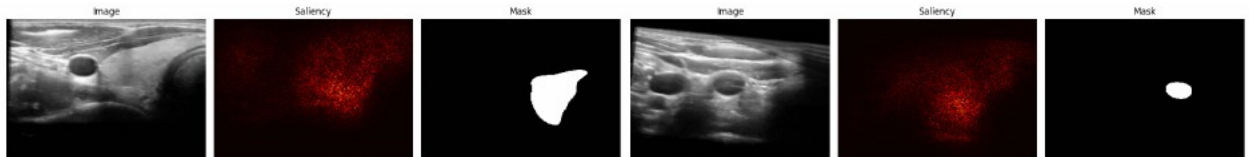


Figure 9: Saliency maps for positive and negative cases

that humans would typically regard as irrelevant regions.

References

Giulia Rubiu, Marco Bologna, Michaela Cellina, Maurizio Cè, Davide Sala, Roberto Pagani, Elisa Mattavelli, Deborah Fazzini, Simona Ibba, Sergio Papa, and Marco Ali. Teeth segmentation in panoramic dental x-ray using mask regional

- convolutional neural network. *Applied Sciences*, 13(13):7947, 2023. doi:10.3390/app13137947
- Yaqi Wang, Fan Ye, Yifei Chen, Chengkai Wang, Chengyu Wu, Feng Xu, Zhean Ma, Yi Liu, Yifan Zhang, Mingguo Cao, and Xiaodiao Chen. A multi-modal dental dataset for semi-supervised deep learning image segmentation. *Scientific Data*, 12(1):117, 2025. doi:10.1038/s41597-024-04306-9
- Blake VanBerlo, Jesse Hoey, and Alexander Wong. A survey of the impact of self-supervised pretraining for diagnostic tasks in medical x-ray, ct, mri, and ultrasound. *BMC Medical Imaging*, 24(1):79, 2024. doi:10.1186/s12880-024-01253-0
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009, June 2022.
- Laith Alzubaidi, Muthana Al-Amidie, Ahmed Al-Asadi, Amjad J Humaidi, Omran Al-Shamma, Mohammed A Fadhel, Jinglan Zhang, Jesus Santamaría, and Ye Duan. Novel transfer learning approach for medical imaging with limited labeled data. *Cancers*, 13(7):1590, 2021.
- Kun Liu, Shuyi Ling, and Sidong Liu. Semi-supervised medical image classification with pseudo labels using coalition similarity training. *Mathematics*, 12(10):1537, 2024.
- Ling-Li Zeng, Kai Gao, Dewen Hu, Zhichao Feng, Chenping Hou, Pengfei Rong, and Wei Wang. Ss-tbn: A semi-supervised tri-branch network for covid-19 screening and lesion segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(8):10427–10442, 2023.
- Banafshe Felfeliyan, Nils D Forkert, Abhilash Hareendranathan, David Cornel, Yuyue Zhou, Gregor Kuntze, Jacob L Jaremko, and Janet L Ronsky. Self-supervised-rcnn for medical image segmentation with limited data annotation. *Computerized Medical Imaging and Graphics*, 109:102297, 2023.
- Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A simple framework for masked image modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9653–9663, June 2022.
- Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. Beit: Bert pre-training of image transformers. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.
- Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. ibot: Image bert pre-training with online tokenizer. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.
- Trinh Le Ba Khanh, Duy-Phuong Dao, Ngoc-Huynh Ho, Hyung-Jeong Yang, Eu-Tteum Baek, Gueesang Lee, Soo-Hyung Kim, and Seok Bong Yoo. Enhancing u-net with spatial-channel attention gate for abnormal tissue segmentation in medical imaging. *Applied Sciences*, 10(17):5729, 2020.
- Zhen Tao, Hua Dang, Yueting Shi, Weijiang Wang, Xiaohua Wang, and Shiwei Ren. Local and context-attention adaptive lca-net for thyroid nodule segmentation in ultrasound images. *Sensors*, 22(16):5984, 2022.